HARDWARE

Cerebras Deep Dive Cerebras 深度解析

Disaggregated Inference, Wafer-Scale Engines, The OpenAI and Amazon Deals, IPO Thoughts

解耦推理、晶圆级引擎、OpenAI 与 Amazon 交易详情、IPO 展望

TECH FUND

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Disaggregated Inference—The Amazon Partnership, Benefits vs Drawbacks

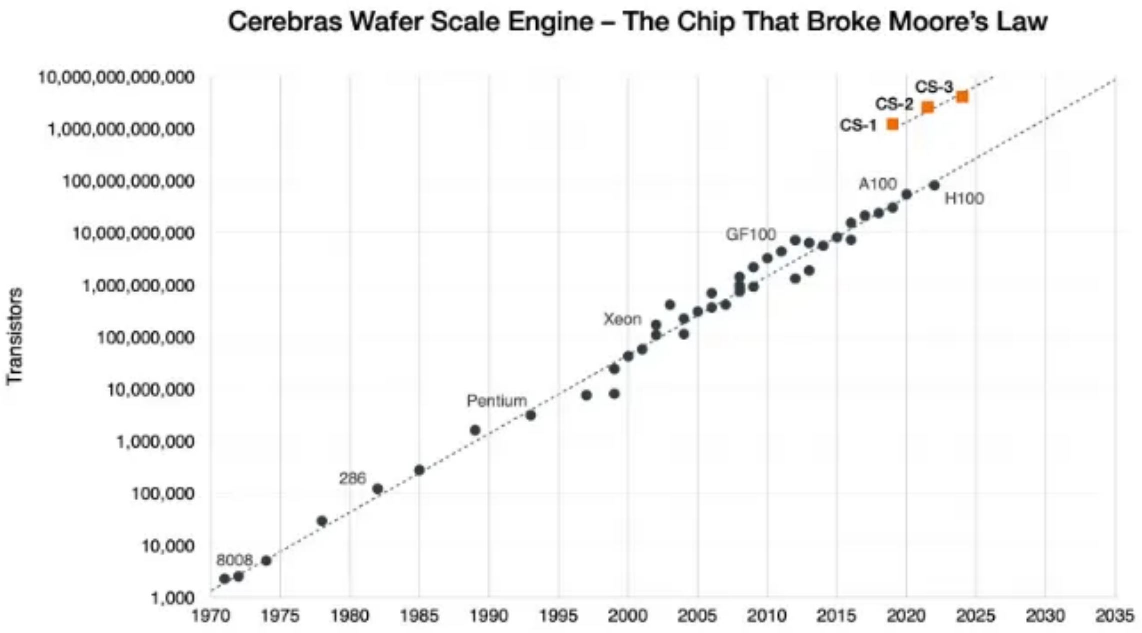
解构式推理——亚马逊合作伙伴关系，利弊分析

Cerebras accelerators have a radically different design than current XPUs. Instead of dicing a 12-inch silicon wafer into dozens of individual chips, Cerebras leaves the wafer intact to create a single, gargantuan processor.

Cerebras 加速器的设计与目前的 XPU 截然不同。Cerebras 并没有将 12 英寸的硅晶圆切割成数十个独立的芯片，而是保留了晶圆的完整性，从而打造出一个巨大的单一处理器。

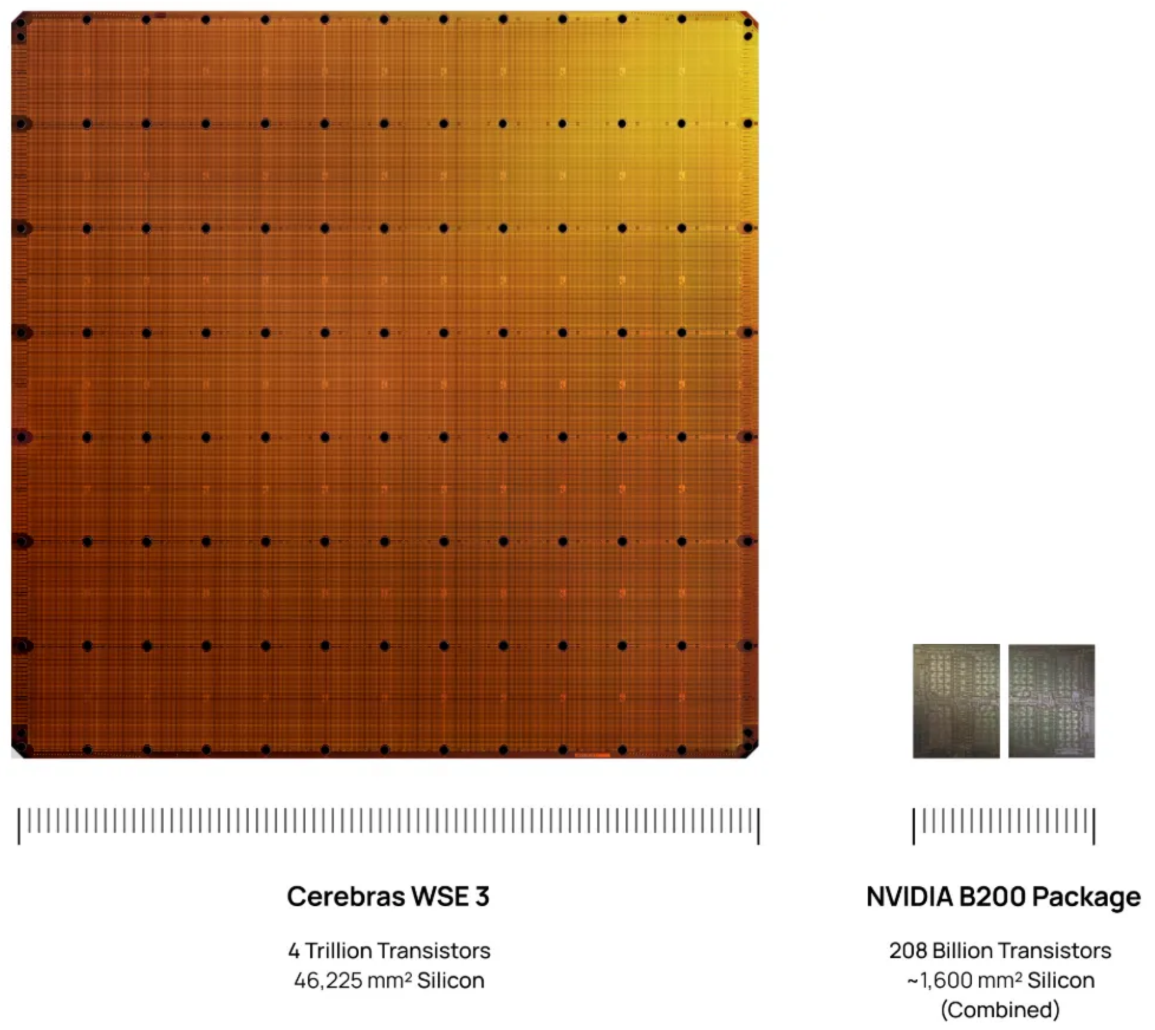
The current iteration, the Wafer-Scale Engine 3, is fabricated by TSMC on a 5nm process featuring 4 trillion transistors distributed across 900,000 AI compute cores.

当前的迭代版本 Wafer-Scale Engine 3 由台积电采用 5nm 工艺制造，在 900,000 个 AI 计算核心中分布着 4 万亿个晶体管。



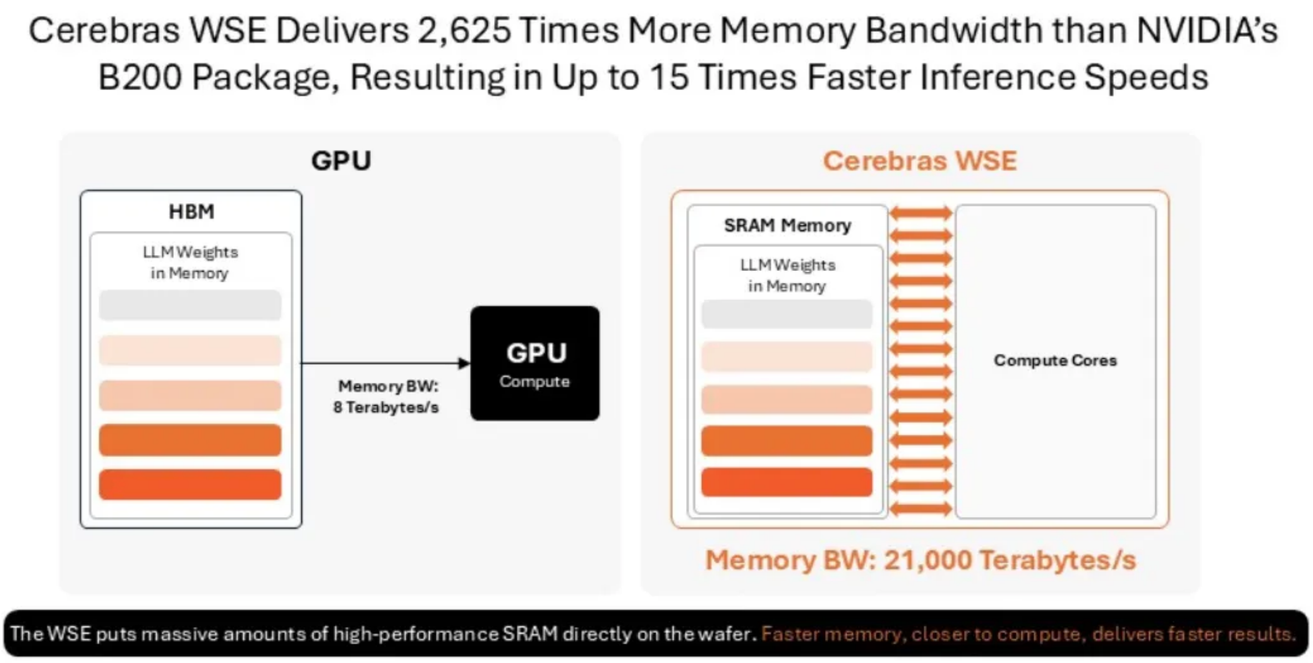
Its wafer scale accelerator is 57 times larger than a single Nvidia B200 die:

其晶圆级加速器的尺寸比单个 Nvidia B200 裸片大 57 倍：



This monolithic design addresses the primary bottleneck in LLM workloads: the memory wall. By keeping 44 GB of SRAM directly on the wafer, Cerebras' WSE delivers a massive 21 petabytes per second of memory bandwidth, resulting into much faster inference (up to 15x according to the company compared to Nvidia B200):

这种单体设计解决了 LLM 工作负载中的主要瓶颈：内存墙。通过在晶圆上直接保留 44 GB 的 SRAM，Cerebras 的 WSE 提供了高达每秒 21 PB 的海量内存带宽，从而实现了更快的推理速度（据该公司称，与 Nvidia B200 相比，速度提升高达 15 倍）：

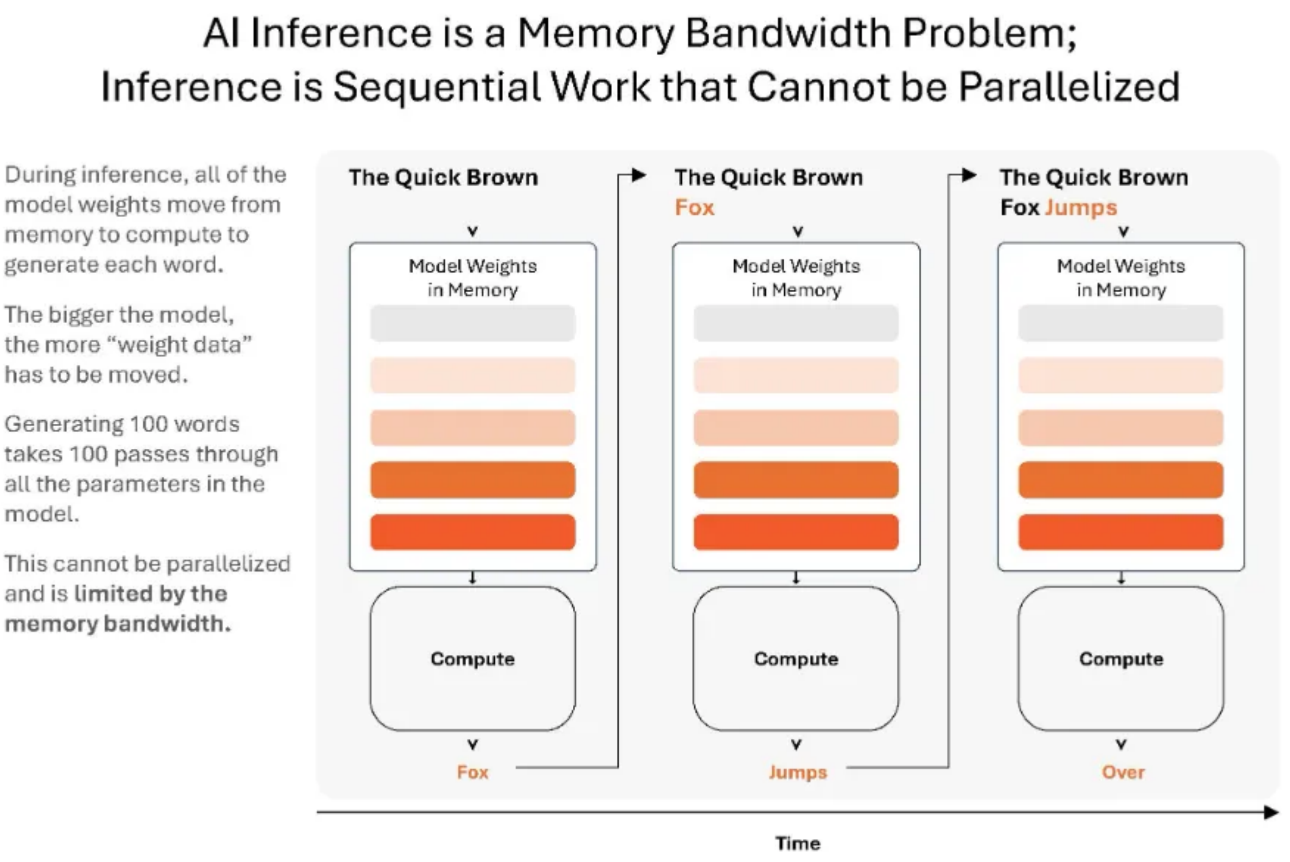


For a GPU to calculate the next token, it needs to multiply the current state by the model's weights. Because of how autoregressive decoding works, you cannot process the next token until the current token is finished. So, to generate one single token, the GPU must read the entire model from its VRAM (HBM) into its compute cores.

对于 GPU 而言，要计算下一个 token，它需要将当前状态与模型权重相乘。由于自回归解码的工作原理，在当前 token 完成之前，无法处理下一个 token。因此，为了生成一个 token，GPU 必须将整个模型从其 VRAM (HBM) 读取到计算核心中。

Because the GPU is doing very little math per byte of data moved, this means decoding is memory-bandwidth bound. Cerebras' design minimizes this bottleneck by storing the model weights in SRAM, which is located right next to the compute cores.

由于 GPU 在移动每字节数据时进行的数学运算非常少，这意味着解码受限于内存带宽。Cerebras 的设计通过将模型权重存储在紧邻计算核心的 SRAM 中，最大限度地减少了这一瓶颈。



Nvidia is aware of the massive bandwidth advantage of SRAM, and so recently announced a unique inference solution that combines the best of both worlds—Rubin for the prefill and the attention phase of decoding (which are compute heavy) and Groq for the final token generation phase of decoding (which is bandwidth intensive).

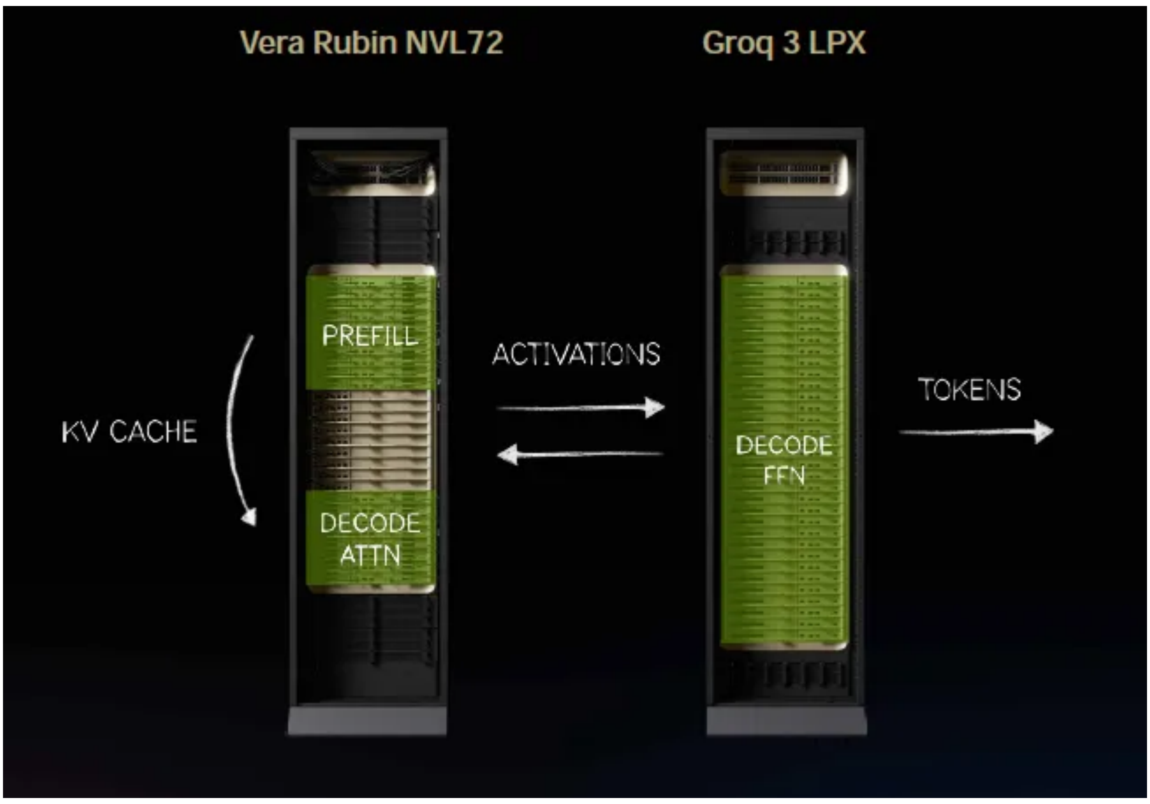
Nvidia 意识到了 SRAM 巨大的带宽优势，因此最近宣布了一种独特的推理解决方案，结合了两者的优点——使用 Rubin 进行预填充（prefill）和解码的注意力阶段（这些是计算密集型的），并使用 Groq 进行解码的最终 token 生成阶段（这是带宽密集型的）。

In the first phase, handled by Rubin, the AI figures out what the prompt and its context (stored in the KV cache) are about.

在第一阶段，由 Rubin 处理，AI 会识别提示词及其上下文（存储在 KV 缓存中）的具体含义。

While in the final phase, handled by Groq, the model applies its learned knowledge (i.e. its parameters or weightings) to the specific query:

而在最终阶段，由 Groq 处理，模型将其学到的知识（即其参数或权重）应用于特定的查询：



A key drawback of SRAM-based inference systems like Groq and Cerebras is that SRAM doesn't have much memory capacity. So, you have to connect 45 Cerebras CS-3 wafers together to hold a 1 trillion parameter model.

基于 SRAM 的推理系统（如 Groq 和 Cerebras）的一个主要缺点是 SRAM 的内存容量不大。因此，必须将 45 个 Cerebras CS-3 晶圆连接在一起，才能容纳一个拥有 1 万亿参数的模型。

Cerebras can technically cluster up to 2,048 wafers together, so even a state-of-the-art 10 trillion parameter model fits well within this theoretical maximum.

从技术上讲，Cerebras 可以将多达 2,048 个晶圆集群在一起，因此即使是拥有 10 万亿参数的最先进模型，也完全处于这一理论最大值的容纳范围之内。

The KV cache stores all the context from the user queries and so takes up a lot of memory. This is why you want to network together an SRAM-based system with a classic XPU server which has loads of HBM, DRAM, and even SSD. As context windows get longer and longer with AI agents doing multi-turn reasoning, the KV cache explodes in size.

KV 缓存存储了来自用户查询的所有上下文，因此会占用大量内存。这就是为什么需要将基于 SRAM 的系统与拥有海量 HBM、DRAM 甚至 SSD 的传统 XPU 服务器联网。随着 AI 智能体进行多轮推理，上下文窗口变得越来越长，KV 缓存的大小也随之爆炸式增长。

It quickly exceeds what fits in HBM, and even DRAM. As having to recompute the

KV cache is compute expensive, this is the reason why Nvidia recently added additional SSD to their servers.

它很快就会超过 HBM 甚至 DRAM 的容量。由于重新计算 KV 缓存的计算成本非常高，这也是英伟达最近在其服务器中增加额外 SSD 的原因。

This is also the reason why pure SSD stocks rallied strongly last year, however, this is a commoditized and competitive industry, so make sure you don't hold these bags for too long.

这也是纯 SSD 概念股去年强劲反弹的原因，然而，这是一个商品化且竞争激烈的行业，所以请确保不要长期持有这些股票。

The second drawback of a Cerebras inference system is the high price tag. While it can hold a 1 trillion parameter model and more, networking a 45 Cerebras CS-3 cluster together can easily cost $100 million or more. For comparison, a single Nvidia GB200 rack can hold more than six of these models in its HBM at a hardware cost of only $3.5 million.

Cerebras 推理系统的第二个缺点是价格高昂。虽然它可以容纳 1 万亿参数甚至更大规模的模型，但将 45 台 Cerebras CS-3 集群联网，成本很容易达到 1 亿美元或更多。相比之下，单台英伟达 GB200 机架的 HBM 就可以容纳超过六个此类模型，而硬件成本仅为 350 万美元。

The advantage of Cerebras is low latency, whereas the advantage of Nvidia is loads of memory. A single Nvidia rack has so much memory it can handle queries from tons of users at the same time, perfect for a chatbot app or a LLM API.

Cerebras 的优势在于低延迟，而 Nvidia 的优势在于海量内存。单个 Nvidia 机架拥有巨大的内存容量，可以同时处理海量用户的查询，非常适合聊天机器人应用或 LLM API。

Cerebras is more like a Formula 1 car that provides ultra low latency inference for a user willing to pay up for it, e.g. Citadel hedge fund or Goldman's trading desk.

Cerebras 更像是一辆一级方程式赛车，为愿意为此支付高溢价的用户（例如 Citadel 对冲基金或高盛的交易部门）提供超低延迟的推理服务。

So, Cerebras has two key problems—firstly, a lack of context memory, and secondly, an astronomic price tag.

因此，Cerebras 面临两个核心问题：第一，缺乏上下文内存；第二，价格极其昂贵。

The first problem, a lack of context memory, can be solved when you combine the Cerebras system in a disaggregated inference solution with a traditional XPU server which does have the memory. As Nvidia has already shown, you get the best of both worlds—loads of memory and fast decode.

第一个问题，即缺乏上下文内存，可以通过将 Cerebras 系统整合进一个解耦推理解决方案中来解决，该方案配合拥有充足内存的传统 XPU 服务器。正如 Nvidia 已经证明的那样，这样可以兼顾两者的优势——海量内存和快速解码。

This is exactly what Cerebras is now doing with their new Amazon AWS partnership:

这正是 Cerebras 目前通过与亚马逊 AWS 建立的新合作伙伴关系所做的事情：

> "We signed a binding term sheet with Amazon Web Services for AWS to become the first hyperscaler to deploy Cerebras systems in its data centers. Deployment in AWS data centers will require us to meet strict standards for performance, scale, and reliability.
>
> **"我们与 Amazon Web Services 签署了一份具有约束力的条款说明书，AWS 将成为首个在其数据中心部署 Cerebras 系统的超大规模云服务商。在 AWS 数据中心进行部署将要求我们满足性能、规模和可靠性方面的严格标准。**
>
> Pursuant to the term sheet, we will create a co-designed, disaggregated inference-serving solution that will integrate AWS Trainium3 chips with Cerebras CS-3 systems, connected via high-bandwidth networking, to partition inference workloads across Trainium3 and CS-3. Each system will perform the type of computation at which it most excels.
>
> **根据该条款说明书，我们将共同设计一种解耦的推理服务解决方案，该方案将通过高带宽网络将 AWS Trainium3 芯片与 Cerebras CS-3 系统集成，从而在 Trainium3 和 CS-3 之间划分推理工作负载。每个系统都将执行其最擅长的计算类型。**
>
> The approach is expected to deliver 5 times more token throughput in the same hardware footprint, at up to 15 times faster speeds compared to leading GPU-based solutions as benchmarked on leading open-source models.
>
> **与领先的基于 GPU 的解决方案相比，在领先的开源模型基准测试中，该方法预计在相同的硬件占用空间内提供 5 倍的 Token 吞吐量，且速度提升高达 15 倍。"**
>
> ---
>
> Inference disaggregation is a technique that separates AI inference into two stages: prompt processing, or "prefill," and output generation, or "decode." These two stages have different computational characteristics. Prefill is natively parallel and requires very little memory bandwidth.

> **推理解耦（Inference disaggregation）是一种将 AI 推理分为两个阶段的技术：提示词处理（即“预填充” prefill）和输出生成（即“解码” decode）。这两个阶段具有不同的计算特性。预填充本质上是并行的，且对内存带宽的需求极低。**
>
> Decode, on the other hand, is inherently serial and memory bandwidth intensive. Decode is typically the bottleneck. It dominates total inference time, and defines the speed of the user experience.
>
> **相比之下，解码本质上是串行的，且属于内存带宽密集型任务。解码通常是性能瓶颈，它占据了总推理时间的大部分，并决定了用户体验的速度。**
>
> Disaggregated inference would allow Cerebras to operate alongside other architectures, serving as the high-performance engine for decode while other systems handle prefill.
>
> **解耦推理将允许 Cerebras 与其他架构协同运行，由 Cerebras 充当解码的高性能引擎，而其他系统则负责处理预填充。**
>
> ”

AWS will make this hyperfast inference available via Amazon Bedrock for a variety of LLMs, including open-source models.

AWS 将通过 Amazon Bedrock 为包括开源模型在内的多种 LLMs 提供这种极速推理服务。

The second problem with Cerebras is the very high price tag. This also becomes more manageable in a disaggregated inference solution as the throughput, i.e. the number of jobs and users that the Cerebras servers can handle, gets a large boost as Cerebras' wafers can now purely focus on serial token generation.

Cerebras 的第二个问题是其高昂的价格。在解构式推理方案中，这一问题也变得更易于处理，因为吞吐量（即 Cerebras 服务器可以处理的任务和用户数量）得到了大幅提升，因为 Cerebras 的晶圆现在可以纯粹专注于串行 Token 生成。

If those users are willing to pay a substantial premium for these hyperfast tokens, the investment starts to make sense.

如果这些用户愿意为这些超高速 Token 支付高额溢价，那么这项投资就开始变得合理。

This is what Jensen said on the demand for hyperfast inference at GTC:

关于对超高速推理的需求，Jensen 在 GTC 上是这样说的：

> “At the most valuable tier, we're now going to increase performance by 35x, and

> I would add Groq to maybe 25% of my total data center. With the rest of my data center all Vera Rubin."
>
> **"在最有价值的层级，我们现在将性能提高 35 倍，我可能会将 Groq 添加到我数据中心总量的 25% 左右。而我数据中心的其余部分将全部采用 Vera Rubin。"**

So, Jensen has probably provided the biggest bull argument for the TAM Cerebras will be competing in. The second bull argument is that Cerebras looks to be a strong player in this TAM, as the company recently disclosed another interesting partnership…

黄仁勋可能为 Cerebras 将要竞争的潜在市场总量（TAM）提供了最有力的看涨论据。第二个看涨论据是，Cerebras 似乎是该市场中的强有力参与者，因为该公司最近披露了另一项引人注目的合作……

The OpenAI Deal 与 OpenAI 的交易

> "We signed the MRA with OpenAI on December 24, 2025. On January 23, 2026, we began delivering capacity to OpenAI, and on February 12, 2026, OpenAI's Codex-Spark model, powered by Cerebras infrastructure, was made available to the public. Spark is OpenAI's model designed for real-time coding.
>
> **"我们于 2025 年 12 月 24 日与 OpenAI 签署了主收入协议（MRA）。2026 年 1 月 23 日，我们开始向 OpenAI 交付算力；2026 年 2 月 12 日，由 Cerebras 基础设施驱动的 OpenAI Codex-Spark 模型正式向公众开放。Spark 是 OpenAI 专为实时编程设计的模型。**
>
> Using Cerebras, OpenAI's customers can translate ideas into working software in seconds, enabling developers to create software at the speed of thought. OpenAI has committed to purchase 750MW of Cerebras inference compute capacity over the next three years.
>
> **通过使用 Cerebras，OpenAI 的客户可以在几秒钟内将想法转化为可运行的软件，使开发者能够以思维的速度创造软件。OpenAI 已承诺在未来三年内购买 750MW 的 Cerebras 推理计算能力。**
>
> "

This 750MW deal is worth $20 billion, and so spread over three years, this would be around $6.7 billion in annual revenue for Cerebras. We can work out that OpenAI is basically paying here roughly $27 billion per GW of Cerebras capacity. There is an option to expand the agreement to 2GW by 2030, meaning that an additional 1.

这项 750MW 的交易价值 200 亿美元，分三年执行，这意味着 Cerebras 每年将获得约 67 亿美元的收入。我们可以计算出，OpenAI 在此支付的费用大约为每 GW Cerebras 容量 270 亿美元。该协议还包含一个到 2030 年将规模扩大至 2GW 的选项，这意味着 Cerebras 可以在 29 年或 30 年再向 OpenAI 提供 1.25GW 的容量。

25GW can be provided by Cerebras to OpenAI in ‘29 or ‘30. This would mean another $33 billion potential order from OpenAI. OpenAI and Cerebras also agreed to co-design future OpenAI models for future Cerebras hardware.

这将意味着来自 OpenAI 的另一笔价值 330 亿美元的潜在订单。OpenAI 和 Cerebras 还同意针对未来的 Cerebras 硬件共同设计未来的 OpenAI 模型。

OpenAI won’t purchase the equipment. Cerebras will provide the capacity in a managed compute-as-a-service type agreement. As Cerebras has to order the required wafers from TSMC, OpenAI also gave Cerebras a $1 billion working capital loan. If Cerebras can deliver the compute successfully, no interest has to be paid.

OpenAI 不会购买设备。Cerebras 将以托管计算即服务（compute-as-a-service）的协议形式提供容量。由于 Cerebras 必须向台积电（TSMC）订购所需的晶圆，OpenAI 还向 Cerebras 提供了 10 亿美元的营运资金贷款。如果 Cerebras 能够成功交付算力，则无需支付利息。

Otherwise the interest rate is 6% on the loan.

否则，该贷款的利率为 6%。

Cerebras also issued a warrant to OpenAI for 33.5 million Cerebras shares, with vesting directly tied to equipment purchase milestones. So, if OpenAI orders the full 2GW of compute by 2030, they’ll be given the full 33.5 million Cerebras shares.

Cerebras 还向 OpenAI 发行了 3350 万股 Cerebras 股票的认股权证，其归属与设备采购里程碑直接挂钩。因此，如果 OpenAI 在 2030 年前订购满 2GW 的算力，他们将获得全部 3350 万股 Cerebras 股份。

As the exercise price per share is basically $0, OpenAI has a financial incentive to order more Cerebras compute.

由于每股行权价基本为 0 美元，OpenAI 在财务上有动力订购更多 Cerebras 的算力。

The dilution, if this warrant gets fully exercised, will likely be around 10%.

如果该认股权证被完全行使，稀释比例可能在 10% 左右。

Bears will point to the circular financing nature of the agreement, however, we argue that for a startup it's not easy at all to get a hyperscaler or leading AI lab to run inference on your silicon.

看空者会指出该协议具有循环融资的性质，然而我们认为，对于一家初创公司来说，要让超大规模云服务商或顶尖 AI 实验室在你的芯片上运行推理，绝非易事。

A large AI player will have to make considerable investment to deploy models on your hardware—compilers will have to be written, the equipment has to be tested and verified, software layers have to be coded to automatically deploy models on a variety of underlying hardware, etc. All of this costs considerable engineering hours.

大型 AI 厂商必须投入大量资源才能在你的硬件上部署模型——需要编写编译器、测试和验证设备、编写软件层以在各种底层硬件上自动部署模型等等。所有这些都需要耗费大量的工程工时。

In the meantime, AI is progressing at breakneck speed. While AI labs will want to optimize their hardware, their key focus is on not falling behind in this race. The value of their business will be determined by capturing sustainable market share and revenues. For example, by winning enterprise AI workloads.

与此同时，AI 领域正以惊人的速度发展。虽然 AI 实验室希望优化其硬件，但他们的核心重点是在这场竞赛中不掉队。其业务价值将取决于能否夺取可持续的市场份额和收入。例如，通过赢得企业级 AI 工作负载。

In order to do this, you have to train differentiated models, like Claude Code. Improving margins by optimizing hardware will be a secondary objective.

为了实现这一目标，你必须训练具有差异化的模型，例如 Claude Code。通过优化硬件来提高利润率将是次要目标。

Also on an accounting basis, the value of these warrants will be booked as contra revenues in the coming years. From a US GAAP perspective, these warrants are treated as a sales discount, which lowers the net revenue Cerebras ultimately recognizes from OpenAI.

此外，从会计角度来看，这些认股权证的价值将在未来几年计入冲减收入（contra revenues）。根据美国通用会计准则（US GAAP），这些认股权证被视为销售折扣，这将降低 Cerebras 最终从 OpenAI 确认的净收入。

Business Background & Founder's Letter

业务背景与创始人致辞

Cerebras does not sell standalone silicon—it packages its wafer into a full system which integrates power delivery, liquid cooling and networking; occupying one-third of the space of a standard data center rack. Revenue so far has primarily been driven by sovereign and university supercomputers in Abu Dhabi.

Cerebras 并不销售独立的芯片——它将其晶圆封装成一个完整的系统，集成了供电、液冷和网络功能；仅占用标准数据中心机架三分之一的空间。到目前为止，其收入主要由阿布扎比的主权国家级和大学超级计算机驱动。

To make it easy to deploy AI models on Cerebras wafers, the company provides a software platform which includes a Pytorch compiler, an inference stack with APIs, and a cluster management system providing scheduling, telemetry, and health monitoring at scale.

为了让在 Cerebras 晶圆上部署 AI 模型变得简单，该公司提供了一个软件平台，其中包括 Pytorch 编译器、带有 API 的推理栈，以及一个提供大规模调度、遥测和健康监测的集群管理系统。

This is even integrated with the most popular IDE, VS Code, so you can write an AI model with Pytorch in VS Code and then compile it to run on Cerebras.

它甚至与最流行的 IDE —— VS Code 进行了集成，因此你可以在 VS Code 中使用 Pytorch 编写 AI 模型，然后将其编译并在 Cerebras 上运行。

The management team, led by CEO Andrew Feldman, brings a proven track record. Cerebras' founders previously established SeaMicro—a pioneer in high-density and energy-efficient microservers—which was acquired by AMD for $334 million in 2012. Feldman subsequently served two years as VP at AMD, after which he co-founded Cerebras.

由首席执行官 Andrew Feldman 领导的管理团队拥有卓越的往绩。Cerebras 的创始人此前创立了 SeaMicro——高密度和节能微型服务器领域的先驱——该公司于 2012 年被 AMD 以 3.34 亿美元收购。Feldman 随后在 AMD 担任了两年的副总裁，之后共同创立了 Cerebras。

More than 50 engineers from SeaMicro joined Cerebras, highlighting that Feldman can create a cohesive engineering culture.

超过 50 名来自 SeaMicro 的工程师加入了 Cerebras，这凸显了 Feldman 能够建立具有凝聚力的工程文化。

CTO and co-founder Sean Lie has a longer background at AMD and was also an engineer at SeaMicro:

首席技术官兼联合创始人 Sean Lie 在 AMD 拥有更深厚的背景，且同样曾是 SeaMicro 的工程师：

> "Sean Lie is one of our co-founders and has served in various roles since 2016,

including most recently as our Chief Technology Officer since April 2022. From April 2012 to June 2015, Mr. Lie served as Chief Architect, Data Center Server Solutions at AMD. From March 2008 to March 2012, Mr.

"Sean Lie 是我们的联合创始人之一，自 2016 年以来担任过多个职位，最近自 2022 年 4 月起担任我们的首席技术官。从 2012 年 4 月到 2015 年 6 月，Lie 先生在 AMD 担任数据中心服务器解决方案首席架构师。从 2008 年 3 月到 2012 年 3 月，Lie 先生

Lie served as Lead Hardware Architect of the IO virtualization fabric ASIC at SeaMicro. From July 2004 to February 2008, Mr. Lie worked as a microprocessor architect at AMD in the advanced architecture team. Mr. Lie holds an M.Eng. and a B.S. in Electrical Engineering and Computer Science from the Massachusetts Institute of Technology.

Lie 曾担任 SeaMicro 公司 IO 虚拟化架构 ASIC 的首席硬件架构师。2004 年 7 月至 2008 年 2 月，Lie 先生在 AMD 的高级架构团队担任微处理器架构师。Lie 先生拥有麻省理工学院（MIT）电气工程与计算机科学专业的硕士（M.Eng.）和学士（B.S.）学位。

"

It is notoriously difficult to get high yields with wafer-scale silicon—a single defect can ruin an entire wafer. Cerebras circumvents this by outfitting the hardware with millions of redundant compute and memory cells, bypassing defective areas dynamically at the manufacturing level.

众所周知，晶圆级硅片的良率极难保证——单个缺陷就可能毁掉整个晶圆。Cerebras 通过在硬件中配置数百万个冗余计算和存储单元来规避这一问题，在制造层面动态绕过缺陷区域。

While this solves the yield problem, the resulting systems are extremely capital intensive. A single physical node is estimated to cost between $2-3 million and can draw up to 27kW of power.

虽然这解决了良率问题，但由此产生的系统属于极度资本密集型。据估计，单个物理节点的成本在 200 万至 300 万美元之间，功耗高达 27kW。

CS-3 System Specifications

Processor	1x Cerebras 3rd Gen Wafer-Scale Engine (WSE-3)
Processor Memory	44 GB total, 21 PB/s SRAM bandwidth
Processor Performance	FP16 125 petaFLOPS
I/O Bandwidth	1.2 Tb/s
System Power	27 kW max
System Dimensions	16 RU
Liquid Cooling	Water Filtering: 50um Water Chemistry: PG25 Water Flow Rate: 100L/min/system Water Temperature: 20 ± 2ºC

* FLOPS shown as sparse

From the founder's letter, this gives useful background on how they're thinking about the business and industry:

从创始人致辞中，可以了解到他们对业务和行业的思考背景，这些信息非常有参考价值：

> "The first bet: that existing general-purpose processors would not be sufficient, and that what has always been true throughout the history of compute would also be true for AI – that transformative compute workloads require purpose-built silicon. This is what PCs did for x86, graphics did for GPUs, and mobile did for ARM.
>
> **第一个赌注：现有的通用处理器将不足以应对需求，而且计算历史上始终如一的规律在 AI 领域也将同样适用——即变革性的计算工作负载需要定制化的芯片。正如个人电脑之于 x86，图形处理之于 GPU，以及移动设备之于 ARM。**
>
> The second bet: that modifying existing compute architectures would not realize AI's potential. We would need to build a new computer architecture from first principles, optimized in every way for AI.
>
> **第二个赌注：修改现有的计算架构无法释放 AI 的潜力。我们需要从第一性原理出发，构建一种在各个方面都为 AI 优化的全新计算机架构。**

> At a computational level, AI is a communication-bound problem. The faster compute communicates with memory, and the faster compute communicates with other compute, the faster and smarter the AI, and the better the user experience. The enemy of speed is communication latency.
>
> **在计算层面，AI 是一个受通信限制的问题。计算与内存之间的通信越快，计算与计算之间的通信越快，AI 就会运行得越快、越聪明，用户体验也就越好。速度的敌人是通信延迟。**
>
> And since communication is thousands of times faster on-chip than across chips, the best way to reduce latency is to keep communication on-chip.
>
> **由于片上通信比跨芯片通信快数千倍，降低延迟的最佳方法就是将通信保留在芯片内。**

> Our answer: build the largest commercial chip in the history of the computer industry. We used the entire wafer for one chip: a technique called wafer-scale integration.
>
> **我们的答案是：制造计算机工业历史上最大的商用芯片。我们将整块晶圆用于制造一颗芯片：这项技术被称为晶圆级集成（wafer-scale integration）。**
>
> Wafer-scale integration allowed us to bring together quantities of compute and

memory never before assembled on a single commercial chip and deliver AI at previously unimaginable speeds.

晶圆级集成使我们能够将前所未有的计算量和内存容量汇聚在单颗商用芯片上，并以以往难以想象的速度交付人工智能。

Nobody knew how to yield a chip 58 times larger than the leading GPU. Nobody knew how to deliver power to a chip the size of a dinner plate without melting the motherboard. Nobody knew how to package such a big chip without cracking it.

没有人知道如何让一颗比领先 GPU 大 58 倍的芯片达到良率。没有人知道如何向一颗餐盘大小的芯片供电而不熔化主板。没有人知道如何封装这么大的芯片而不使其破裂。

Nobody knew how to cool a chip of this size, with air or water, without the coolant getting warm before it reached the other side.

没有人知道如何用空气或水为这种尺寸的芯片散热，而不让冷却剂在到达另一端之前就变热。

We delivered our first systems in 2020 and the second generation in 2022. We had built something extraordinary, but the market wasn't ready. A few visionary customers in supercompute and life sciences saw the potential, but to most of the world, the benefits were not immediately apparent. Change arrived with ChatGPT.

我们在 2020 年交付了第一代系统，并在 2022 年推出了第二代。我们打造了非凡的产品，但市场尚未准备好。少数在超级计算和生命科学领域具有远见的客户看到了其潜力，但对世界上大多数人来说，其优势并非显而易见。改变随 ChatGPT 的出现而到来。

Suddenly everyone was talking about AI, and entrepreneurs were extending the art of the possible.

突然之间，每个人都在谈论 AI，创业者们正在不断拓展"可能"的边界。

By early 2025, AI was smart enough to be valuable, and people were using AI everywhere. Inference usage exploded. Suddenly, everyone remembered the lesson from Google search: speed produces more satisfied and more frequent users, while even tiny delays significantly reduce user satisfaction, search frequency, and search revenue.

到 2025 年初，AI 已经足够聪明并产生价值，人们在各个角落使用 AI。推理需求爆发式增长。突然间，每个人都想起了谷歌搜索的教训：速度能带来更满意、更频繁的用户，而哪怕是微小的延迟也会显著降低用户满意度、搜索频率和搜索收入。

The market had shifted. Everyone realized that fast AI is more useful than slow AI.

市场已经发生了转变。每个人都意识到，快速的 AI 比缓慢的 AI 更有用。

By the end of 2025, it was clear: fast inference was powering the highest-value workloads. Fast inference was making engineers more productive because fast AI coding agents could write code, edit it, test it, and get a product to market more quickly.

到 2025 年底，情况已经明朗：快速推理正在为最高价值的工作负载提供动力。快速推理提高了工程师的生产力，因为快速的 AI 编程智能体可以编写、编辑、测试代码，并更快地将产品推向市场。

Fast AI made lawyers, analysts, bankers, doctors, and researchers more productive than their counterparts waiting on slow insights. For many workloads, Cerebras is up to 15 times faster than leading GPU-based solutions as benchmarked on leading open-source models. In some more exotic workloads, we have been more than 1,000 times faster.

快速 AI 让律师、分析师、银行家、医生和研究人员比那些等待缓慢洞察的同行更具生产力。在领先的开源模型基准测试中，对于许多工作负载，Cerebras 的速度比领先的基于 GPU 的解决方案快 15 倍。在某些更特殊的工作负载中，我们的速度甚至快了 1,000 倍以上。

In March 2026, we started a multi-year partnership with AWS to bring fast inference to an even bigger scale through global distribution. This is planned to give every startup, AI native, and enterprise company easy access to Cerebras's blisteringly fast inference.

2026 年 3 月，我们开始与 AWS 建立多年合作伙伴关系，通过全球分发将快速推理推向更大的规模。此举旨在让每家初创公司、AI 原生企业和大型企业都能轻松获得 Cerebras 极速的推理能力。

We were among the first semi-conductor companies to have a cloud business. We deliver hardware on-premises to customers concerned about data security and sovereignty. And we reach other customers through the cloud – both the Cerebras cloud and soon the AWS cloud.

我们是首批拥有云业务的半导体公司之一。我们为关注数据安全和主权的客户提供本地部署的硬件。同时，我们也通过云端触达其他客户——包括 Cerebras 云以及即将推出的 AWS 云。

This allows us to reach customers who want to rent compute by month, year, or to pay by the token, and provides Cerebras with an attractive mix of recurring revenue as well as lumpier hardware revenue.

这使我们能够触及那些希望按月、按年租赁算力或按 token 付费的客户，并为 Cerebras 提供了由经常性收入和波动较大的硬件收入组成的极具吸引力的收入组合。

The fundamental building blocks in computer architecture are calculation (cores), the storage of results (memory), and the movements of results to where they are needed (communication). Our inventions are blurring the lines that have traditionally forced tradeoffs between memory capacity and speed.

计算机架构的基本构建模块是计算（核心）、结果存储（内存）以及将结果移动到所需位置（通信）。我们的发明正在模糊那些传统上迫使人们在内存容量和速度之间进行权衡的界限。

And the massive size of our processor serves as a foundation for pioneering communication techniques that are only available to wafer scale solutions.

我们处理器巨大的尺寸，为开创性的通信技术奠定了基础，而这些技术仅适用于晶圆级解决方案。

”

Financials

The current financials are nothing to get too excited about—the company mainly derives its revenues from AI compute in Abu Dhabi while due to the heavy R&D investment, the business is operating on sizeable operating losses (-40% EBIT margins):

目前的财务状况并不值得过度兴奋——该公司的收入主要来自阿布扎比的 AI 算力业务，同时由于高额的研发投入，业务正处于大幅经营亏损状态（息税前利润率为 -40%）：

	Year Ended December 31, 2025	Year Ended December 31, 2024
	(in thousands)	
Revenue:		
Hardware	$ 358,440	$ 211,965
Cloud and other services	151,551	78,287
Total revenue	509,991	290,252
Cost of revenue(1):		
Hardware	204,746	137,310
Cloud and other services	106,174	30,204
Total cost of revenue	310,920	167,514
Gross profit	199,071	122,738
Operating expenses:		
Research and development(1)	243,319	158,234
Sales and marketing(1)	70,645	20,980
General and administrative(1)	30,969	44,962
Total operating expenses	344,933	224,176
Loss from operations	(145,862)	(101,438)
Other income (expense), net	390,746	(378,237)
Income (loss) before income tax	244,884	(479,675)
Income tax expense	7,057	1,927
Net income (loss)	$ 237,827	$ (481,602)

Around one-third of operating losses are non-cash, but dilutive costs, in the form of share based compensation:

运营亏损中约有三分之一是非现金但具有稀释性的成本，以股权激励支出的形式存在：

	Year Ended December 31, 2025	Year Ended December 31, 2024
	(in thousands)	
GAAP operating loss	$ (145,862)	$ (101,438)
Add: Stock-based compensation expense	49,767	58,564
Non-GAAP operating loss	$ (96,095)	$ (42,874)

As already mentioned, warrants will be deducted from revenues which will give some revenue headwinds:

正如前文所述，认股权证将从营收中扣除，这将带来一定的营收阻力：

> "When we begin to recognize contra-revenue amounts from the warrants in the first quarter of 2026, we expect quarterly revenue growth rates will decline from recent trends."
>
> **"当我们从 2026 年第一季度开始确认认股权证产生的冲减营收金额时，我们预计季度营收增长率将较近期趋势有所下降。"**

Gross margins are fairly low at around 40% but this is still a small scale business. If the company can successfully scale up to provide compute-as-a-service to AWS and OpenAI, long term gross margins should be fairly attractive.

毛利率处于约 40% 的较低水平，但这仍是一个小规模业务。如果公司能够成功扩大规模，为 AWS 和 OpenAI 提供计算即服务（compute-as-a-service），长期毛利率应当会相当可观。

The two partnerships (so far)—with OpenAI and Amazon AWS—are with two of